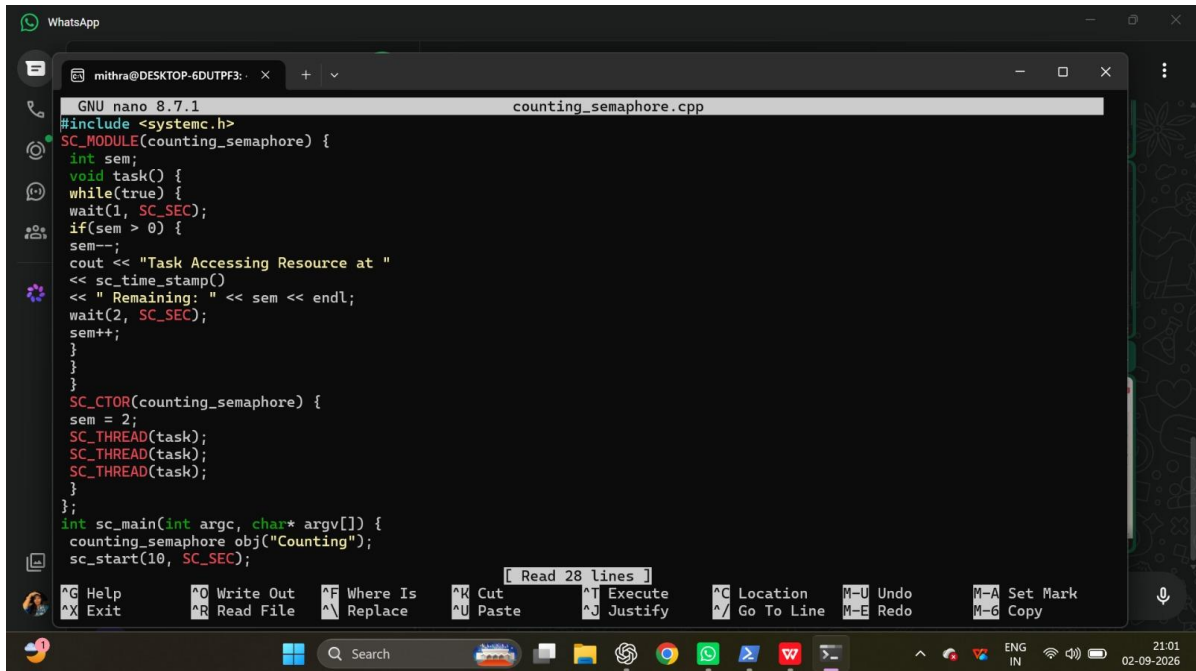


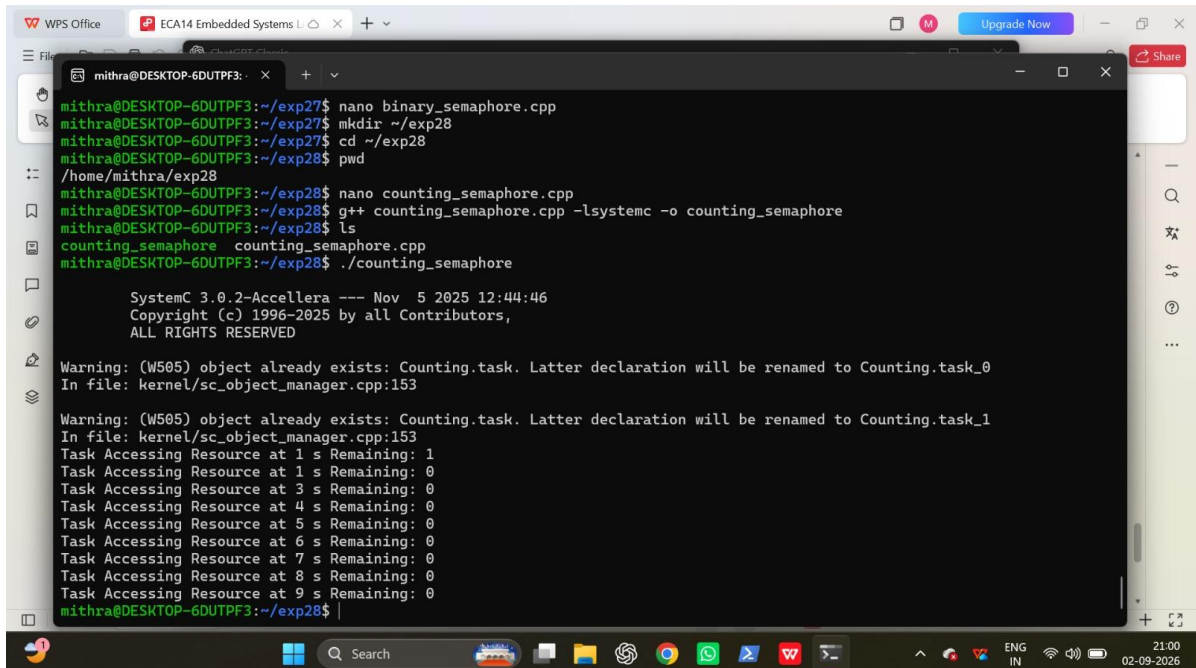
Exp No: 28

COUNTING SEMAPHORE IN SYSTEMC



```
GNU nano 8.7.1 counting_semaphore.cpp
#include <systemc.h>
SC_MODULE(counting_semaphore) {
    int sem;
    void task() {
        while(true) {
            wait(1, SC_SEC);
            if(sem > 0) {
                sem--;
                cout << "Task Accessing Resource at "
                    << sc_time_stamp()
                    << " Remaining: " << sem << endl;
                wait(2, SC_SEC);
                sem++;
            }
        }
    }
    SC_CTOR(counting_semaphore) {
        sem = 2;
        SC_THREAD(task);
        SC_THREAD(task);
        SC_THREAD(task);
    }
};

int sc_main(int argc, char* argv[]) {
    counting_semaphore obj("Counting");
    sc_start(10, SC_SEC);
}
```



```
mithra@DESKTOP-6DUTPF3:~/exp27$ nano binary_semaphore.cpp
mithra@DESKTOP-6DUTPF3:~/exp27$ mkdir ~/exp28
mithra@DESKTOP-6DUTPF3:~/exp27$ cd ~/exp28
mithra@DESKTOP-6DUTPF3:~/exp28$ pwd
/home/mithra/exp28
mithra@DESKTOP-6DUTPF3:~/exp28$ nano counting_semaphore.cpp
mithra@DESKTOP-6DUTPF3:~/exp28$ g++ counting_semaphore.cpp -lsystemc -o counting_semaphore
mithra@DESKTOP-6DUTPF3:~/exp28$ ls
counting_semaphore  counting_semaphore.cpp
mithra@DESKTOP-6DUTPF3:~/exp28$ ./counting_semaphore

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED

Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0
In file: kernel/sc_object_manager.cpp:153
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1
In file: kernel/sc_object_manager.cpp:153
Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0
Task Accessing Resource at 5 s Remaining: 0
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
mithra@DESKTOP-6DUTPF3:~/exp28$
```